

Digital Empowerment Network

TASK-03

Email Spam Classification

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Task: 03(Email Spam Classification)

Email Spam Classification

1. 1. Introduction

The purpose of this project is to build a machine learning model that classifies email messages as either **Spam** or **Ham** (not spam). Spam detection is an important real-world application of Natural Language Processing (NLP) used to filter unwanted or harmful messages.

2. 2. Dataset

- Dataset: spam.csv
- Total Records: ~5,500 messages
- Columns Used:
 - **Category:** Label (ham or spam)
 - **Message:** The text content of the email/SMS message

3. 3. Methodology

The project was completed in three main phases:

4. Data Cleaning & Preprocessing

- Converted all text to lowercase.
- Removed punctuation, numbers, and special characters.
- Tokenized messages into words.

- Removed stopwords using NLTK.
- Applied stemming with Porter Stemmer.
- Joined tokens back into cleaned sentences.

5. Vectorization

- Used **TF-IDF Vectorizer** with n-grams to convert text into numerical form.
- Parameters:
 - lowercase=True
 - ngram_range=(1,2)
 - min_df=1 (to keep rare but important words)

6. Model Training & Evaluation

- Models Used:
 - Multinomial Naive Bayes
 - Logistic Regression
 - Linear Support Vector Machine (SVM)
- Evaluation Metrics:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Confusion Matrix

7. Visualizations

- **Class Distribution:** Showed number of spam vs ham messages in the dataset.
- **Confusion Matrix Heatmap:** Displayed model performance visually.

8. Results

Model	Accuracy	Precision (Spam)	Recall (Spam)	F1-Score (Spam)
Multinomial Naive Bayes	0.971	0.962	0.905	0.933
Logistic Regression	0.982	0.977	0.934	0.955
Linear SVM (Best Model)	0.984	0.981	0.934	0.957

Best Model: Linear SVM – chosen based on overall highest F1-Score.

9. Predictions on Sample Messages

The model was tested with unseen messages. Examples:

Message	Prediction
Congratulations! You have won a \$1000 Walmart gift card. Click here...	Spam
Are we still meeting at 3pm today?	Ham
URGENT: Your account is locked. Verify your details immediately.	Spam
Ok, see you soon!	Ham

WINNER! Free vacation to Bahamas...	Spam
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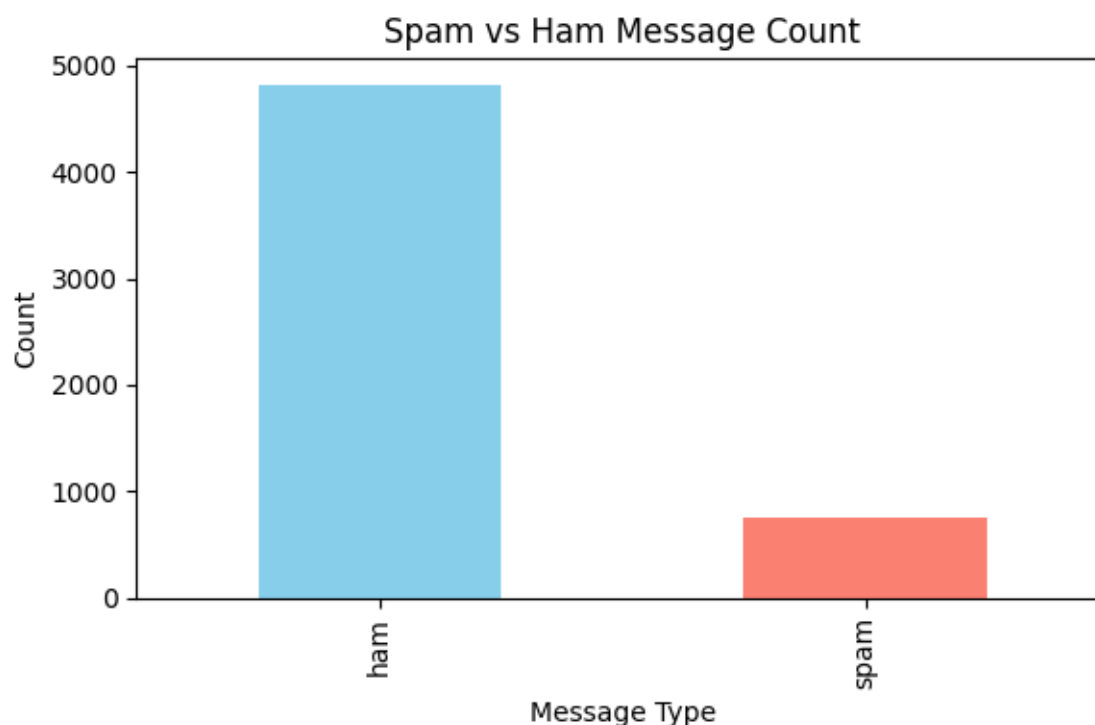
10. Conclusion

The spam classification system successfully distinguishes spam and ham messages with high accuracy. Using TF-IDF with n-grams and the Linear SVM classifier provided the best performance. Future improvements can include:

- Adding more training data.
- Trying deep learning models like LSTM or BERT.
- Using lemmatization instead of stemming.

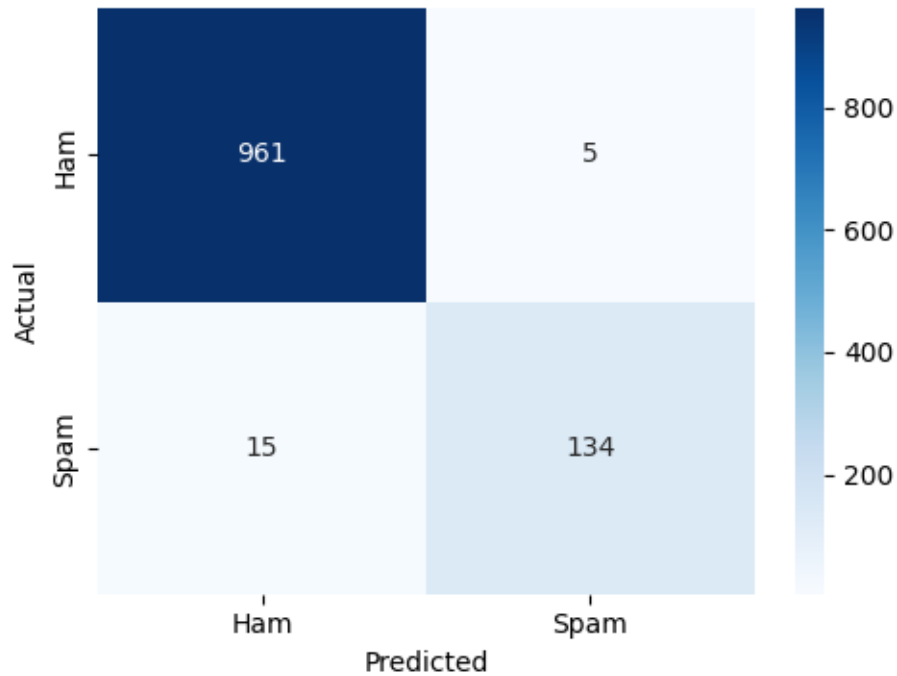
VISUALIZATIONS

Spam vs Ham count



Confusion Matrix for best model

Confusion Matrix - Best Model (Linear SVM)



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